Exp 6(a) Dijkstra's Algorithm

Command Prompt

```
C:\DSA>gcc exp6a.c -o exp6a.exe

C:\DSA>exp6a.exe
Enter number of vertices: 4
Enter cost matrix (999 = no edge):
0 4 1 999
4 0 2 5
1 2 0 8
999 5 8 0
Enter source vertex: 0

Shortest distances from vertex 0:
Vertex 0 : 0
Vertex 1 : 3
Vertex 2 : 1
Vertex 3 : 8

C:\DSA>_
```

Exp 6(b) Floyd-Warshall Algorithm

Command Prompt

```
C:\DSA>gcc exp6b.c -o exp6b.exe

C:\DSA>exp6b.exe
Enter number of vertices: 4
Enter cost matrix (999 = no edge):
0 4 1 999
4 0 2 5
1 2 0 8
999 5 8 0

All-pairs shortest path matrix:
0       3       1       8
3       0       2       5
1       2       0       7
8       5       7       0

C:\DSA>_
```

Exp 7(a) Kruskal's Algorithm

Command Prompt

```
C:\DSA>gcc exp7a.c -o exp7a.exe

C:\DSA>exp7a.exe
Enter number of vertices: 4
Enter cost matrix (0 = no edge):
0 2 3 0
2 0 1 4
3 1 0 5
0 4 5 0

Edges in MST:
Edge 1: (1 - 2) cost = 1
Edge 2: (0 - 1) cost = 2
Edge 3: (1 - 3) cost = 4
Minimum cost = 7

C:\DSA>_
```

Exp 7(b) Prim's Algorithm

Command Prompt

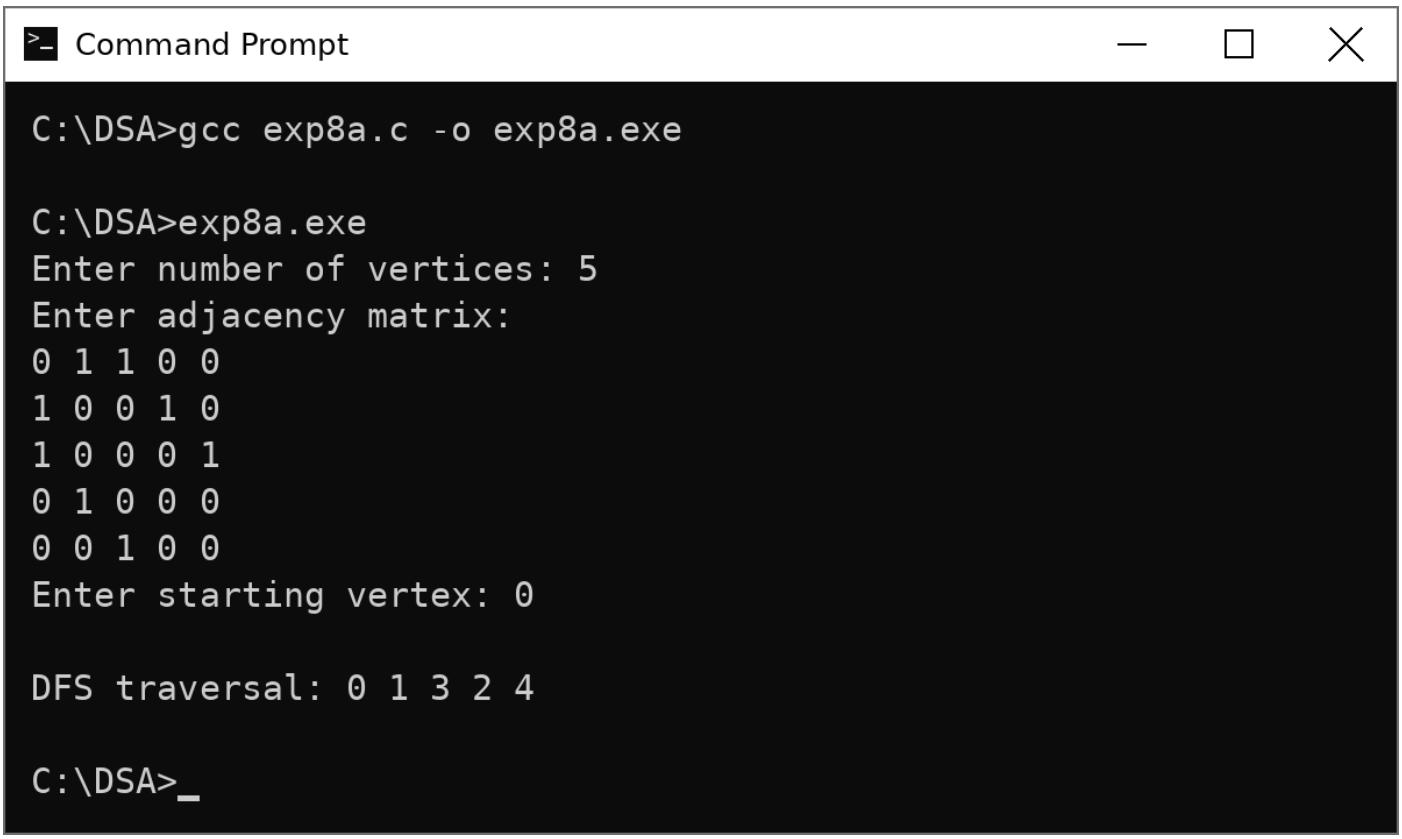
```
C:\DSA>gcc exp7b.c -o exp7b.exe

C:\DSA>exp7b.exe
Enter number of vertices: 4
Enter cost matrix (0 = no edge):
0 2 3 0
2 0 1 4
3 1 0 5
0 4 5 0

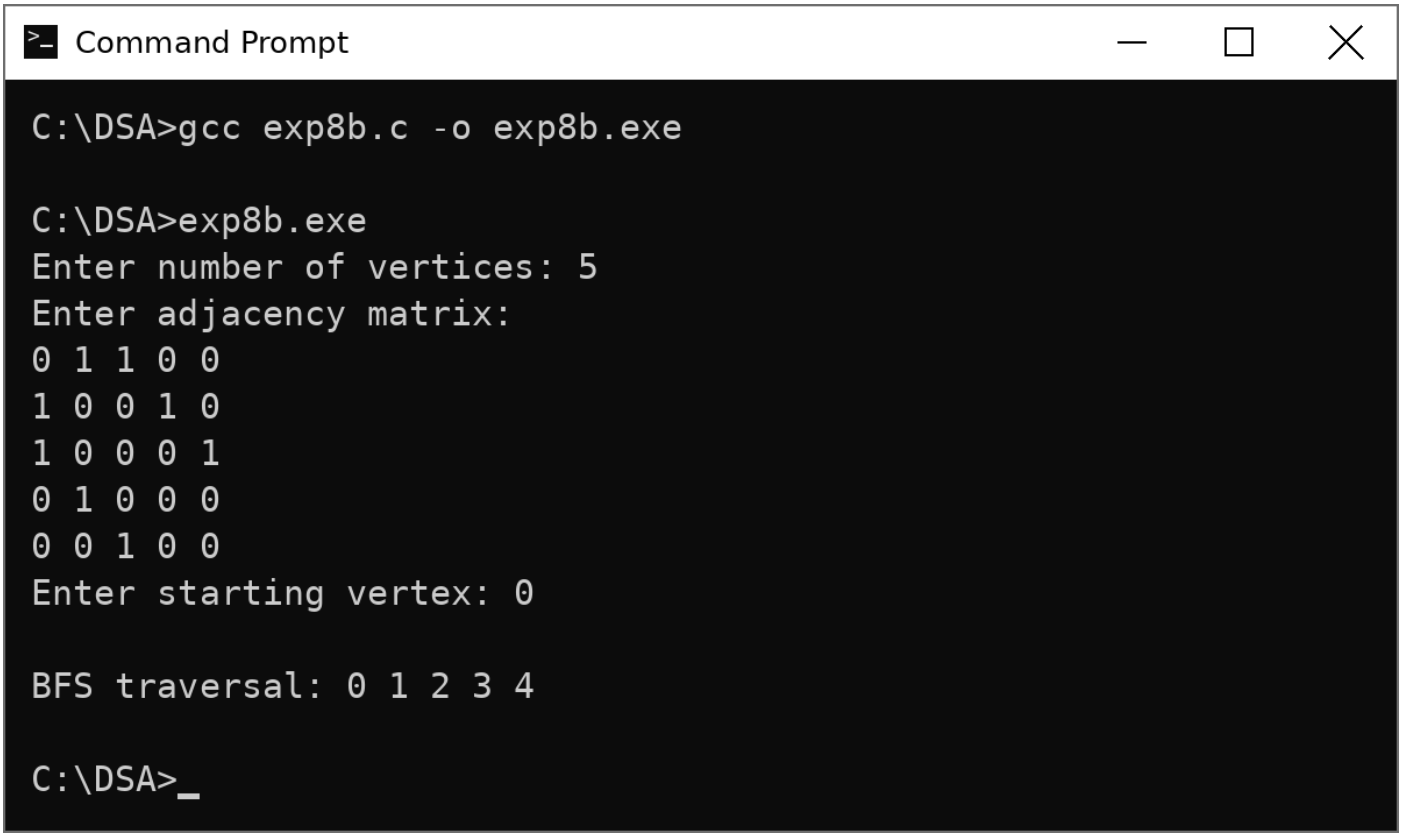
Edges in MST:
Edge 1: (0 - 1) cost = 2
Edge 2: (1 - 2) cost = 1
Edge 3: (1 - 3) cost = 4
Minimum cost = 7

C:\DSA>_
```

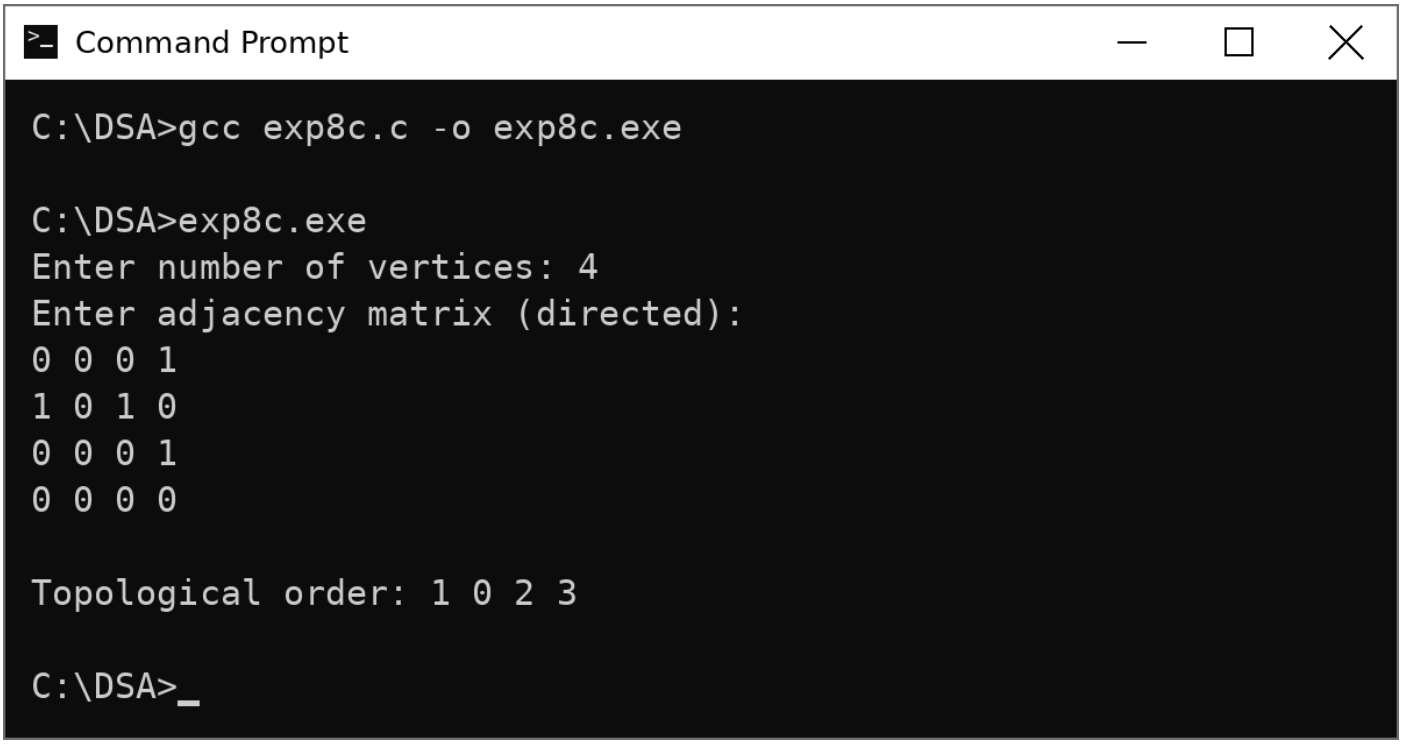
Exp 8(a) Depth First Search (DFS)



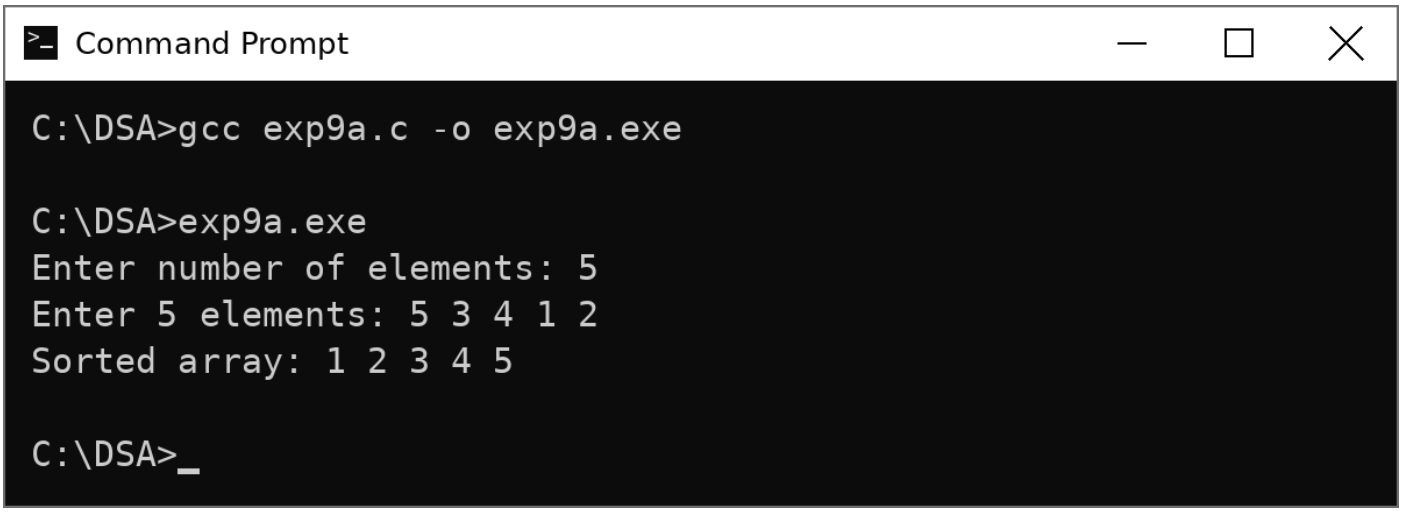
Exp 8(b) Breadth First Search (BFS)



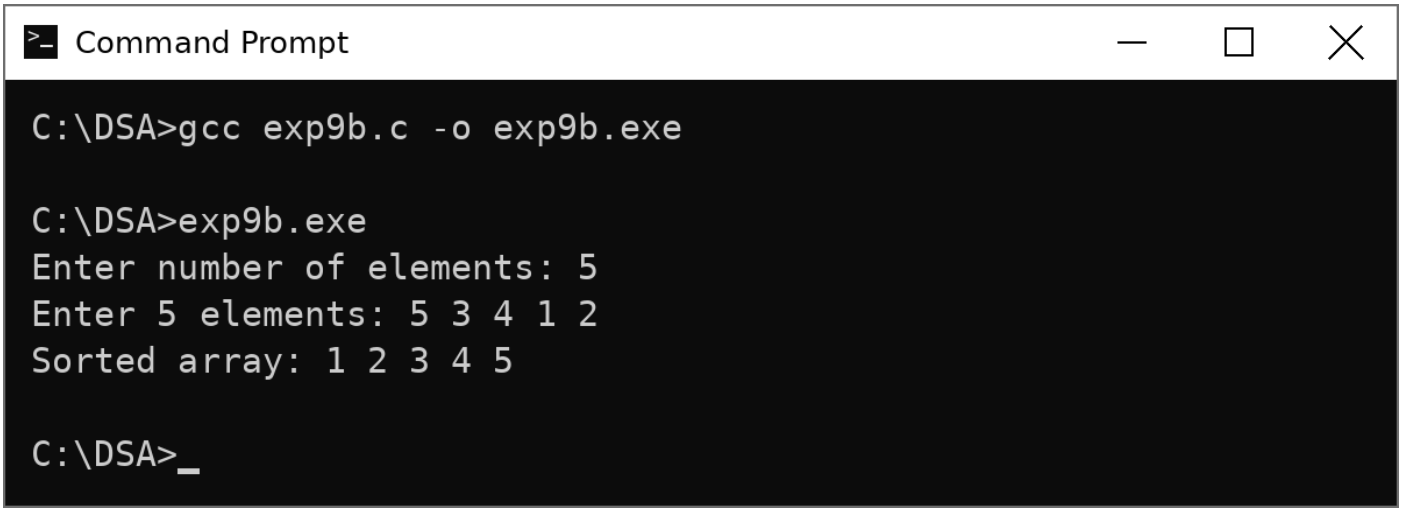
Exp 8(c) Topological Sorting



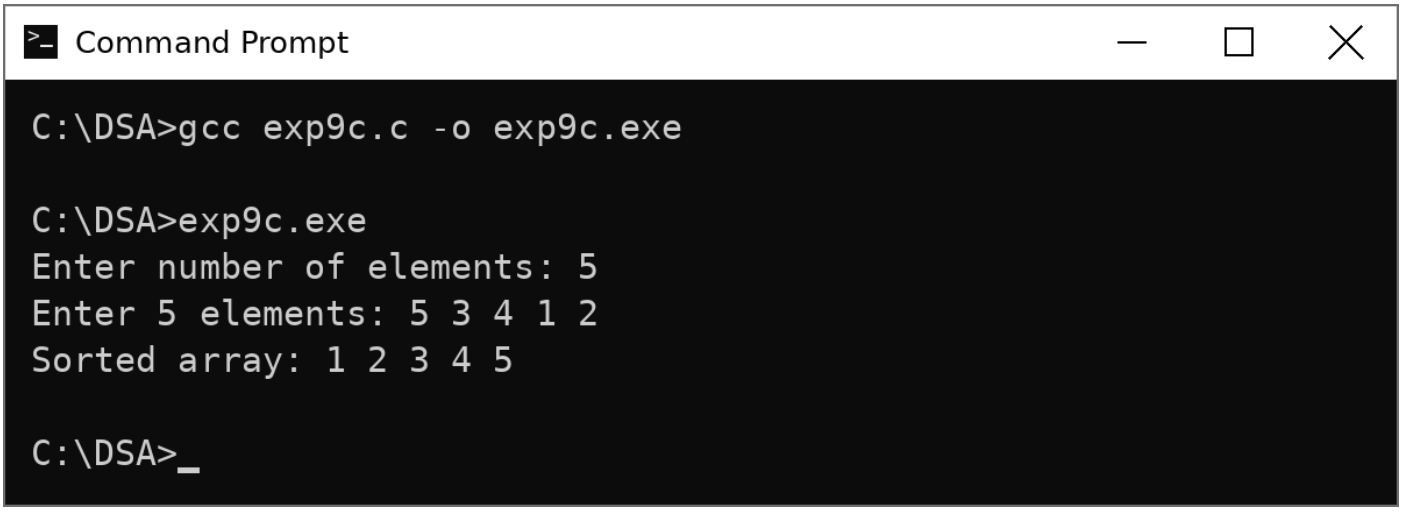
Exp 9(a) Insertion Sort



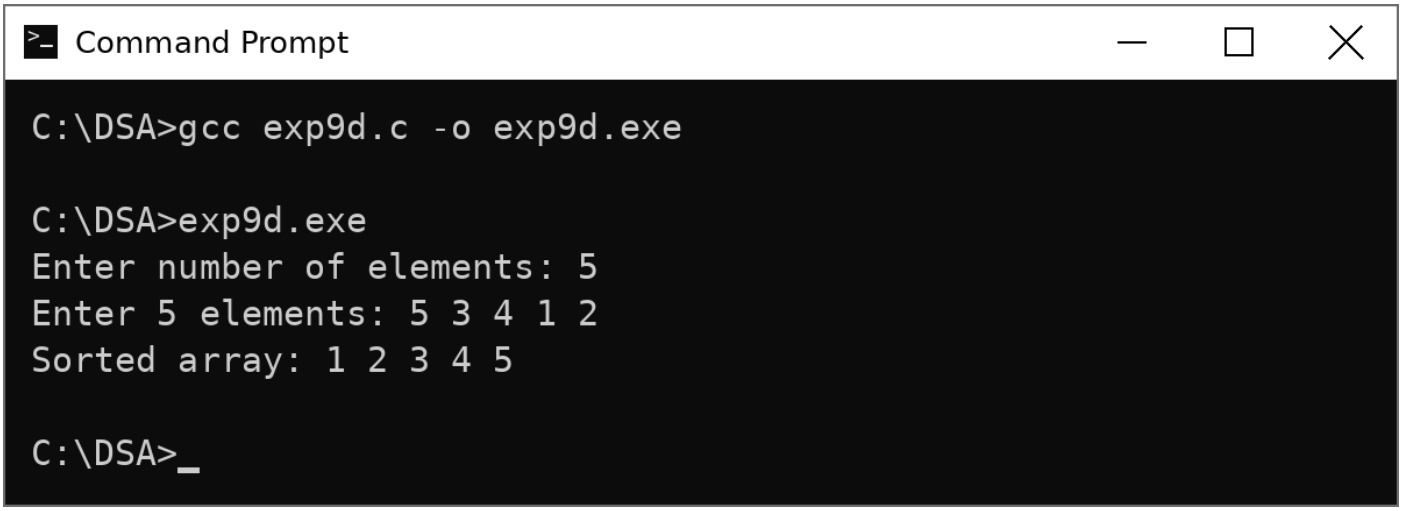
Exp 9(b) Selection Sort



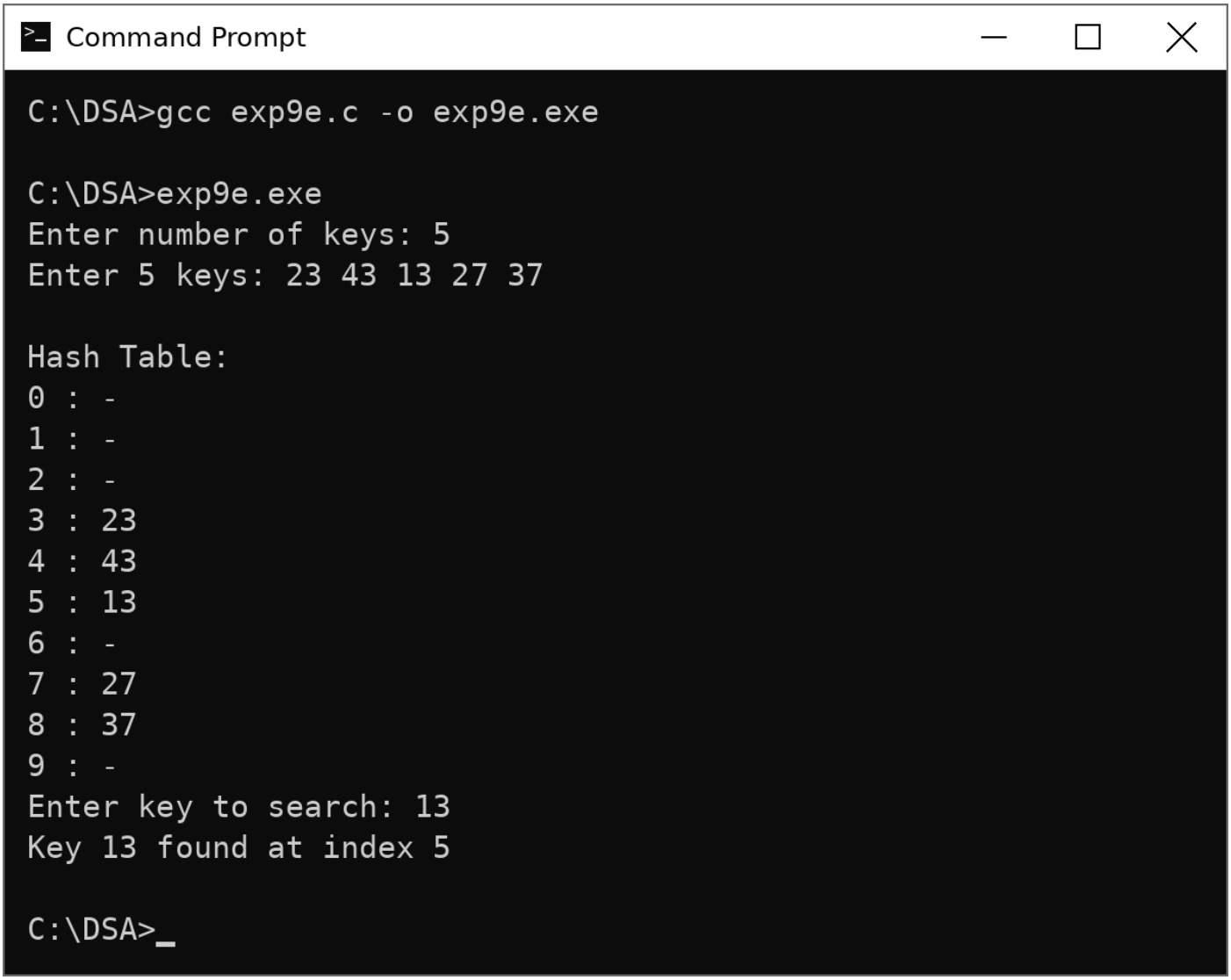
Exp 9(c) Quick Sort



Exp 9(d) Merge Sort



Exp 9(e) Hash Table - Linear Probing



Exp 9(f) Hash Table - Quadratic Probing

